

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 22

AIR POLLUTION - 9

NATIONAL AIR QUALITY INDEX (AQI)

- Launched by the Ministry of Environment, Forests and Climate Change in April 2015. It is an initiative under 'Swachh Bharat' program.
- It helps the common man to judge the air quality within his vicinity.
- While the earlier measuring index was limited to three indicators, the current measurement index had been made quite comprehensive by the addition of more parameters.



Ministry of Environment, Forest
& Climate Change



WHY IS THE NEW AQI NECESSARY?

- Quality of data from some cities remains weak and the standards set for pollutants fall short of World Health Organization recommendations.
- The pollution related analysis using vast number of complex parameters of the old index was complicated for the common man to understand.
- The new AQI considers eight pollutants (PM₁₀, PM_{2.5}, Nitrogen Oxides, Sulphur Di Oxide, Carbon Monoxide, Ozone, ammonia, and lead).



CATEGORIES OF AIR POLLUTION UNDER AQI

- There are six AQI categories.
- They are namely
 - ✓ Good
 - ✓ Satisfactory
 - ✓ Moderately polluted
 - ✓ Poor
 - ✓ Very Poor and
 - ✓ Severe.

AQI	Remark	Color Code	Possible Health Impacts
0-50	Good		Minimal impact
51-100	Satisfactory		Minor breathing discomfort to sensitive people
101-200	Moderate		Breathing discomfort to the people with lungs, asthma and heart diseases
201-300	Poor		Breathing discomfort to most people on prolonged exposure
301-400	Very Poor		Respiratory illness on prolonged exposure
401-500	Severe		Affects healthy people and seriously impacts those with existing diseases

NATIONAL CLEAN AIR PROGRAMME (NCAP)

- NCAP is a five-year action plan with a target of 20-30% reduction in concentrations of PM₁₀ and PM_{2.5} by 2024, with 2017 as the base year.
- NCAP includes preparing city specific action plans for the 102 non-attainment cities, identified by CPCB on the basis of their ambient air quality data between 2011 and 2015.
- **The programme also aims:**
 - ✓ To expand the national air quality monitoring network.
 - ✓ To build capacity for air pollution management
 - ✓ To raise public awareness about the hazards of air pollution



NCAP INITIATIVES

- The National Air Quality Monitoring Network will be augmented.
- Air Quality Management Plan for the cities chosen.
- Indoor Air Pollution Monitoring & Management
- National Emission Inventory – this is an inventory of the quantity of pollutants discharged into the air.
- Network of Technical Institutions
- International cooperation
- At the national level, the implementation will be done by an apex committee at the Environment Ministry level. At the state level, committees at the Chief Secretary level will oversee the implementation of the scheme.



NGT'S DIRECTIONS

- The timeline to reduce air pollution by 20-30% by 2024 needs to be reduced.
- The target of reduction needs to be increased.
- It suggested the Ministry to act on - shift to e-vehicles and CNG vehicles, intensifying public transport system, mechanical cleaning of roads, enhancement of public parking facilities, improvement in fuel quality, and traffic management.
- It directed the state pollution control boards to ensure the assessment and installation of AAQMs within 6 months.



- In 2015 CPCB identified 102 non-attainment cities across the country, which are cities that fail to meet the National Ambient Air Quality Standards.
- Around 76.8% Indians breathe air that is worse than the levels recommended by the National Ambient Air Quality Standards.
- The WHO's air quality database updated in 2018, lists cities on the basis of their annual average particulate matter, PM 2.5 concentration. According to this list, 14 of the world's 20 most polluted cities are in India.
- Industrialization and Urbanization have resulted in deterioration of India's air quality.
- Of the millions of premature deaths in the world that occur each year due to outdoor air pollution, the highest number of deaths occurs in India.
- According to the Lancet report published in December 2018, in 2017 alone, close to 1.24 million people died in India due to air pollution related health problems.
- 51.4% of the people who died due to air pollution were younger than 70 years.
- An average Indian will live 1.7 years longer if the air pollution is less.



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